

Question			Marking details		Marks available					Prac
					AO1	AO2	AO3	Total	Maths	
4	(a)	(i)		Test wire and reference wire made from the same material or identical (1) Incorrect to refer to common support Temperature change will have the same effect on both wires (1) Don't accept reference to being at the same temperature			2	2		2
		(ii)		Extension / point beyond which the wire will not return to its original length / permanently deformed (1) Don't accept reference to Hooke's law or limit of proportionality <u>Removing load</u> (or equivalent) and observe whether or not wire returns to original length or when the load is removed the extension values are the same (1)	1		1	2		1
		(iii)	I	Improve accuracy by reducing [fractional] uncertainty or to provide a measurable extension / longer wire or produces greater extension / more accurate extension Don't accept reduce uncertainty only or a wider range of results			1	1		1
			II	Improve accuracy (or reduce uncertainty) in cross-sectional area / to obtain a mean value for diameter / check for uniformity			1	1		1
	(b)	(i)		$A = 8.04 \times 10^{-8} \text{ [m}^2\text{]} / 0.08 \text{ mm}^2$ (1) (or by implication) Load extension combination e.g. 28 N, $4.8 \times 10^{-3} \text{ m}$ (1) (or by implication) e.g. 5833.3 Substitution into $E = \frac{Fl}{Ae}$ e.g. $\frac{28 \times 2.4}{8.04 \times 10^{-8} \times 4.8 \times 10^{-3}}$ (1) (ecf on A and load extension combination) $E = 1.74 \times 10^{11} \text{ N m}^{-2}$ or Pa or sensible alternative (1) unit mark Accept 2 or 3 sig figs	1	1 1		4	4	

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		(ii)		Energy stored = $\frac{1}{2} \times 2.4 \times 10^{-3} \times 14$ [substituting values from the graph] (1) Energy = 16.8×10^{-3} [J] (1) Accept 17×10^{-3} [J] Alternative: $E = \frac{1}{2} kx^2 = \frac{1}{2} \times 5\,833.3 \times (2.4 \times 10^{-3})^2$ (1) Energy = 16.8×10^{-3} [J] (1) Accept 17×10^{-3} [J] Deduct 1 mark for factor of 10 slip		2		2		
				Question 4 total	2	5	5	12	4	5